P=Pi-Cam <-vectors

If your point relative to the camera is

$$P = (x, y, z)$$

and you apply **pitch first** by angle ϕ:

$$x_1 = x$$

$$y_1 = y\cos\phi - z\sin\phi$$

$$z_1 = y\sin\phi + z\cos\phi$$

Then apply **yaw** by angle θ:

$$x_2 = x_1\cos\theta - z_1\sin\theta$$

$$y_2 = y_1$$

$$z_2 = x_1\sin\theta + z_1\cos\theta$$

Then project:

$$x_{2D} = \frac{x_2}{z_2}$$

$$y_{2D} = \frac{y_2}{z_2}$$

So your pipeline could be:

$$\boxed{\text{subtract camera position} \rightarrow \text{pitch} \rightarrow \text{yaw} \rightarrow \text{projection}}$$

This is supposed to be the matrix that does all this

If we combine camera translation + rotation first, we get:

$$\begin{bmatrix} c_\theta & -s_\theta s_\phi & -s_\theta c_\phi & A \\ 0 & c_\phi & -s_\phi & B \\ s_\theta & c_\theta s_\phi & c_\theta c_\phi & C \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where:

$$A = -c_\theta C_x + s_\theta s_\phi C_y + s_\theta c_\phi C_z$$

$$B = -c_\phi C_y + s_\phi C_z$$

$$C = -s_\theta C_x - c_\theta s_\phi C_y - c_\theta c_\phi C_z$$

Then applying our simple projection produces:

$$M = \begin{bmatrix} f_x c_\theta & -f_x s_\theta s_\phi & -f_x s_\theta c_\phi & f_x A \\ 0 & f_y c_\phi & -f_y s_\phi & f_y B \\ s_\theta & c_\theta s_\phi & c_\theta c_\phi & C \\ s_\theta & c_\theta s_\phi & c_\theta c_\phi & C \end{bmatrix}$$

Y and x 2d still have to be found morally for last step

3D object vertex

↓

move/rotate object optional at first

↓

world-space vertex

↓

subtract camera position

↓

undo camera pitch

↓

undo camera yaw

↓

camera-space vertex

↓

near-plane check

↓

perspective projection

↓

normalized 2D coordinates

↓

convert to 640×480 pixel coordinates

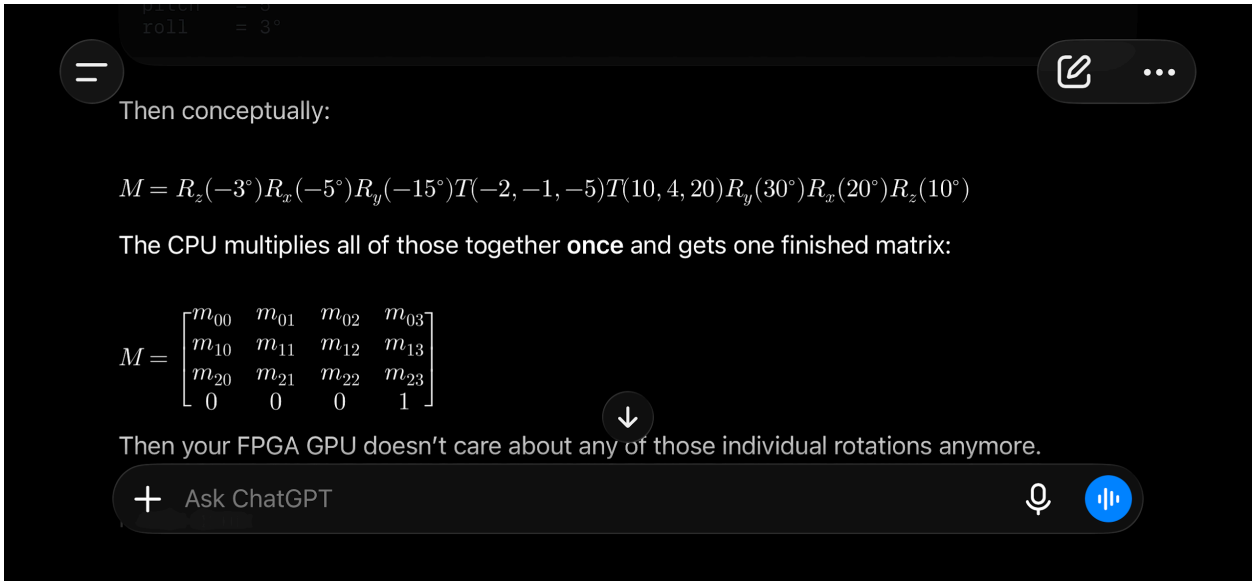
↓

3 projected vertices

↓
triangle setup
↓
rasterization
↓
depth test
↓
choose pixel color
↓
write framebuffer
↓
display

Object:
position = (10, 4, 20)
yaw = 30°
pitch = 20°
roll = 10°

Camera:
position = (2, 1, 5)
yaw = 15°
pitch = 5°
roll = 3°



R_x: rotation around X = pitch

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi & 0 \\ 0 & \sin\phi & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This changes mainly y and

R_y: rotation around Y = yaw

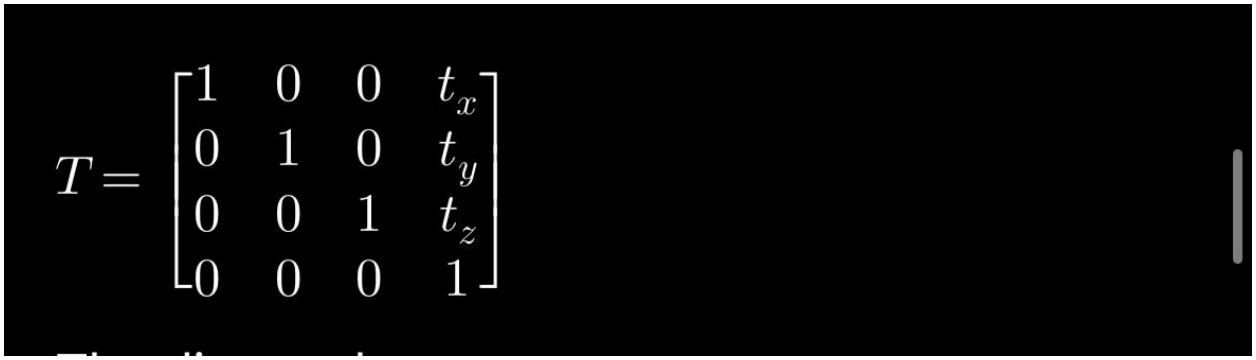
$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This changes mainly x and z.

R_z: rotation around Z = roll

$$R_z(\rho) = \begin{bmatrix} \cos\rho & -\sin\rho & 0 & 0 \\ \sin\rho & \cos\rho & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This changes mainly x and y.



t is just tranfrom